COMMERCIAL ITEM DESCRIPTION

FORMING MACHINE, SHEET METAL, ROTARY, MANUALLY OPERATED, BENCH MOUNTED COMBINATION TYPE

The General Services Administration has authorized the use of this commercial item description in preference to type V of Military Specification MIL-F-23243A.

1. SCOPE

1.1 Scope. This description covers manually operated, bench mounted, sheet metal, combination forming machines of the standard, heavy duty, commercial type that are used in a production environment to perform turning, burring, wiring, edging, flattening, or flanging operations when used with the applicable forming rolls. These combination forming machines are intended for use in sheet metal shops and repair units.

1.2 Classification. The combination forming machines shall be of the following styles and sizes. The style and size to be furnished shall be as specified (see 6.1).

Style A - Regular throat.
Size 18 - 18 gauge (0.0478 inch) capacity.
Size 22 - 22 gauge (0.0299 inch) capacity.
Size 24 - 24 gauge (0.0239 inch) capacity.

Style B - Deep throat.
Size 22 - 22 gauge (0.0299 inch) capacity.
Size 24 - 24 gauge (0.0239 inch) capacity.

2. APPLICABLE DOCUMENTS

2.1 Documents. The following documents of the issue in effect on the date of invitation for bids or request for proposals form a part of this description to the extent specified herein.

Military Specification MIL-M-18058	- Machinery, Metal and Woodworking, Packaging of.
Federal Standard FED-STD-376	- Preferred Metric Units for General Use by the Federal Government.
American Society for Testing and Materials ASTM D 3951	- Standard Practice for Commercial Packaging.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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3. SALIENT CHARACTERISTICS

3.1 Design. The combination forming machine shall be new and of the manufacturer's standard design in effect on the date of the solicitation. The machine shall be capable of forming operations of the rated thickness capacity specified in table 1 for the size machine specified.

TABLE I. Styles A and B combination forming machine, size and characteristics.

	Size	Capacity in mild steel (inches)	Maximum wire diameter (inches)	Distance between shaft centers (inches)	Throat depth (inches)	Gear ratio
Style A	18	0.0478	5/16	2-3/4		2:1
	22	0.0299	5/16	2		1:1
	24	0.0239	3/16	2		1:1
Style B	22	0.0299	1/4	1	6-1/4	1:1
	24	0.0239	3/16	1-7/8	6-1/4	1:1

NOTE: Unless otherwise specified, value shall be not less than the value specified.

3.2 Frame. The frame shall be constructed of steel or cast iron capable of supporting the shafts and drive mechanism and shall provide a suitable base connection for mounting.

3.3 Shafts. The shafts shall be of high grade machine finished steel. The shafts shall be driven by hand crank and gears and shall be fitted to bearings to provide smooth and easy operation at rated capacity.

3.4 Bearings. The bearings shall conform to standard commercial bearings used in machines furnished to the industrial market.

3.5 Gears. The gears shall be machine cut steel or cast iron and shall be totally enclosed or guarded to protect the operator.

3.6 Rolls. The rolls shall be machine finished steel and shall be of sufficient hardness to perform the operations for which designed. Each machine shall be furnished with a set of rolls. When specified (see 6.1), extra rolls shall be furnished as described.

3.6.1 Roll adjustment. Fast adjustment of top roll and angular wheel shall be provided to move from first operation to second operation. The upper roll shall be provided with a quick acting screw adjustment to provide the necessary pressure to the rolls to perform the forming operations.

3.7 Hand crank. The hand crank shall be of sufficient length to ensure operation of the machine without undue strain to the operator.

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3.8 Stand. Unless otherwise specified (see 6.1), each combination forming machine shall be provided with a clamp-on offset stand for bench mounting. The stand offset shall be shaped to bring the machine forward to provide ease of operation.

3.9 Measurement system. All dimensions in this description are given in US units. These dimensions may be converted to SI units through the use of conversion factors and methods specified in FED-STD-376.

3.10 Regulatory requirements. The offeror/contractor is encouraged to use recovered materials to the maximum extent practical.

3.11 Workmanship. The workmanship of the combination forming machine shall be of a quality equal to that of comparable products furnished to the commercial market.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection, examination, and test requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections, examinations, or tests set forth in this description where such inspections, examinations, and tests are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Examination. Each combination forming machine shall be visually examined to determine conformance with all requirements of this description.

4.3 Operational. Each combination forming machine shall be operated without load to ensure proper operation of all moving parts and adjusting mechanisms.

4.4 Packaging inspection. Packaging shall be inspected for conformance with section 5.

5. PACKAGING

5.1 Preservation, packing, and marking. Unless otherwise specified in the contract or purchase order, preservation, packing, and marking shall be in accordance with ASTM D 3951. When specified (see 6.1), level A preservation, level A or level B packing, and marking shall be accomplished in accordance with MIL-M-18058.

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6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

a. Title, number, and date of this commercial item description.
b. Style and size of combination forming machine required (see 1.2).
c. Specify and describe extra rolls, if required (see 3.6).
d. Type of stand, if different (see 3.8).
e. Packaging in accordance with MIL-M-18058, if required specify level of preservation and packing (see 5.1).

6.2 Cross reference data. Items covered by Military Specification MIL-F-23243A are cross referenced to items covered by this description as follows:

MIL-F-23243A	A-A-51172
Type V	Style A (New style) Size 18 (New size) Size 22 (New size) Size 24 (New size) Style B (New style) Size 22 (New size) Size 24 (New size)

MILITARY INTERESTS:

Custodians

Army - AL
Air Force - 99

Review Activities

Air Force - 84
DLA - GS

User Activity

DLA - GS

CIVIL AGENCY COORDIATING ACTIVITY:

GSA - FSS

PREPARING ACTIVITY:

DLA - IP

Project 3441-0131

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
A-A-51172	Forming Machine, Sheet Metal, Rotary, Manually Operated, Bench Mounted Combination Type

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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